

R documentation

of 'README.Rd'

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PDACMOC

PDACMolecularOmniClassifier

Description

This package classifies tumor samples according to several molecular classifiers available using different Machine Learning (ML) approaches. It classifies both tumor and stroma fractions. To classify stroma compartment, it first makes a virtual microdissection through the **ADVOCATE** package.

See 'NEWS.md' for the changes in each version.

Classifiers available for tumor fractions:

1. Collisson et al., 2011.
2. Moffitt et al., 2015.
3. Bailey et al., 2016.
4. Puleo et al., 2018.
5. Chan-Seng-Yue et al., 2020.
6. PDAConsensus (following the methodology of Kamoun et al., 2020).

Classifiers available for stroma fractions:

1. Moffitt et al., 2016.
2. Maurer et al., 2019.
3. PDAConsensus (following the methodology of Kamoun et al., 2020).

Details

The package has been built for Linux (Ubuntu 22.04.3 LTS).

It contains the following files:

- install: conda environment and installer (see Installation).
- inst:
 - examples: it contains a R script ('example.R') with some examples.
 - extdata: it contains the 'Certificate.pdf'.

- gene_signatures: it contains a file with all the final genes included.
- graphs_and_tables: it contains two figures that appear in the Shiny app.
- logos: it contains some institutional logos.
- models: it contains ML models saved as pickle format.
- packages: it contains a compressed file ('ADVOCATE_0.1.0.1.tar.gz') used to install ADVOCATE package.
- saved_workspaces: it contains the workspaces generated with the example script.
- training_data: it contains several files used by the classifiers.
- man: it contains the R markdown files produced automatically by devtools::document() using roxygen2 comments.
- R: it contains the different functions of the package.
- DESCRIPTION: description file.
- LICENSE: license file.
- NAMESPACE: namespace file produced automatically by devtools::document() using roxygen2 comments.
- PDACMOC.Rproj: R project file.
- README.Rd: readme R markdown file.
- README.pdf: readme pdf file.
- .Rproj.user: directory generated by RStudio that stores user-specific configurations and session states.
- .Rbuildignore: file used by R to exclude specific files and directories from package building.
- .git: directory for Git control version.
- .gitattributes: file used to manage attributes for paths in the repository, such as configuration for Git LFS.
- .gitignore: file used by Git to exclude specific files and directories from tracking and version control.

Installation

PDACMOC runs on Linux and needs conda (Miniforge or Miniconda, <https://conda-forge.org/download/>). The installer creates a conda environment called pdacmoc with R, Python and all the dependencies:

```
wget https://github.com/pavillos/PDACMOC/releases/download/v.2.6.0/PDACMOC_2.6.0.tar.gz
tar -xzf PDACMOC_2.6.0.tar.gz
bash PDACMOC/install/install.sh PDACMOC_2.6.0.tar.gz
```

To use it, run conda activate pdacmoc, open R and load the package, pointing **reticulate** to the Python of the environment (the installer prints its path):

```
reticulate::use_python('~/.miniconda3/envs/pdacmoc/bin/python', required = TRUE)
library(PDACMOC)
```

The environment pins scikit-learn 1.3.1 and org.Hs.eg.db 3.18.0, the versions the published models were built with: newer versions change the classification. Do not update them by hand.

Classifying your own files from R

`read.counts()` reads a file of raw counts the same way the Shiny app does: it accepts .tsv, .csv and .txt files, detects the separator and averages rows with duplicated gene IDs.

```
library(PDACMOC)
counts <- read.counts('my_counts.csv')
classification <- omni.classify(counts, gene_id = 'EnsemblID')
```

Shiny app instructions

Use it online at <https://pdacmoc.cnio.es/>, or on your computer with `runPDACMOC()`.

1. Upload a .tsv or .csv file of raw counts with genes in rows and samples in columns (max. 1GB).
2. Choose batch correction, the gene ID type, the tumor classifiers and, optionally, the stroma classifiers.
3. Press 'Run classification'. The Run card shows the progress, the estimated time and any message about your file; you can cancel at any time.
4. When it finishes, the app opens the results. Press 'New classification' to classify another file.

After the process you will find the following:

1. In the Results tab, one table per classifier with the predicted subtype (in the colours used in the paper), its probability and a flag when the probability is below the threshold of that classifier (low confidence). *PDAConsensus* tables also show the NonClassicalScore or ActivatedECM-Score.
2. In the Results tab, the table of tumor/stroma proportions (only with stroma classification). 'E' stands for epithelium, 'S' stands for stroma, and 'O' stands for others. 'conf' refers to 95% confidence intervals.
3. In the Summary tab, the number of samples assigned to each subtype for every classifier.
4. In the Performance tab, the published balanced accuracy of every classifier.

Keywords

PDAC, consensus molecular classifier, transcriptomics, Machine Learning, translational medicine, personalized medicine.

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Examples

```
library(PDACMOC)

file1 <- system.file('examples', 'example.R', package = 'PDACMOC')
dir <- file.path(dirname(file1), '../saved_workspaces/example.RData')
#load(dir)
rm(file1)

# change path to your miniconda3 environment
reticulate::use_python('~/.miniconda3/envs/pdacmoc/bin/python', required = TRUE)

file2 <- system.file('training_data', 'all_datasets_corrected.csv', package = 'PDACMOC')

samples <- read.csv(file2, row.names = 1, check.names = FALSE)
rm(file2)

new_samples <- PDACMOC:::import.and.normalize(samples, batch = FALSE, gene_id = 'EnsemblID')

results_collisson <- PDACMOC:::collisson.classify(new_samples)

results_moffitt <- PDACMOC:::moffitt.classify(new_samples)

results_bailey <- PDACMOC:::bailey.classify(new_samples)

results_puleo <- PDACMOC:::puleo.classify(new_samples)

results_chan <- PDACMOC:::chan.classify(new_samples)

results_consensus <- PDACMOC:::PDAConsensus.classify(new_samples)

vm_result <- PDACMOC:::virtual.microdissect(new_samples)

results_stroma_moffitt <- PDACMOC:::stroma.moffitt.classify(vm_result$vm_S)

results_stroma_maurer <- PDACMOC:::stroma.maurer.classify(vm_result$vm_S)

results_stroma_consensus <- PDACMOC:::stroma.PDAConsensus.classify(vm_result$vm_S)

classification_tumor <- PDACMOC:::omni.classify(samples, batch = FALSE, gene_id = 'EnsemblID',
  classifier = c('Collisson', 'Moffitt', 'Bailey',
    'Puleo', 'Chan-Seng-Yue', 'PDAConsensus'))

classification_all <- PDACMOC:::omni.classify(samples, batch = FALSE, gene_id = 'EnsemblID',
  classifier = c('Collisson', 'Moffitt', 'Bailey',
    'Puleo', 'Chan-Seng-Yue', 'PDAConsensus'),
  stroma = TRUE,
  stroma_classifier = c('Moffitt',
    'Maurer',
    'PDAConsensus'))
```

```
# change path to your browser
options(browser = '/usr/bin/firefox')
shinyjs::useShinyjs()
runPDACMOC()

#save.image(dir)
```

Index

PDACMOC, [1](#)